

Exploring AI-Generated Expert Models: Instructor Interaction and Learner Perceptions in a Physics Class

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Abstract: Generative Artificial Intelligence (GenAI) offers opportunities for automatically creating expert model summaries, though creating them is demanding, requiring significant time and effort. Using a case study, we explored three types of expert model summaries, generated by human, AI, and human-AI collaboration. A new, AI-augmented expert modeling tool, which supports summarization activities, was deployed. Human-AI interaction, linguistic differences among the summaries, and learners' perceptions were explored. Data sources included instructor log data, linguistic measures, and a student evaluation survey. Results showed that students preferred AI-generated summaries for their clarity, depth, and coherence, while human-AI collaboration achieved a balance between fluency and accuracy. Findings signify GenAI's potential to reduce workload but underscore the need to address its limitations.

Introduction

Summarization is an instructional method that helps students “write to learn” and serves as a way to assess their understanding (Kim et al., 2020). Due to their pedagogical value, automatic summary evaluation tools (ASEs) have been implemented, including Graphical Interface of Knowledge Structure (GISK; Kim, 2019) and Student Mental Model Analyzer for Research and Teaching (SMART; Kim, 2021). These tools require a human-made reference summary that serves as a guide for students. However, crafting these expert summaries demands significant time and effort from instructors to produce high-quality content (Kim et al., 2016). Writing effective summaries involves the expertise to extract key ideas, identify core themes, and concisely convey these within a limited word count (Kintsch & van Dijk, 1978), requiring a high level of proficiency. GenAI offers a promising solution by automating the creation of expert model summaries. GenAI can reference specific learning materials to capture essential ideas and build broader connections. A GenAI-enabled expert modeling tool as part of SMART technology was developed. SMART is a web-based ASE designed to help students engage with learning materials through summarization supported by AI-augmented formative feedback (Kim et al., 2023a). The expert modeling tool adopts a human-in-the-loop approach, where human experts can enhance AI-generated texts by infusing their knowledge to co-create an expert model—an expert summary highlighting key concepts. This approach addresses current limitations of AI, such as hallucinations—producing plausible but incorrect information (Bender et al., 2021). This study explored three research questions: (1) How do human experts interact with AI to enhance expert model summaries? (2) How do expert model summaries generated by humans alone, AI alone, and human-AI collaboration differ? and (3) How do learners perceive the quality of the three types of expert models in terms of their preference? Human-AI interaction logs over seven consecutive expert models demonstrated by a physics instructor, accompanied by his reflections were analyzed. A case study was conducted by selecting a learning topic from which three distinct expert model summaries were created. These summaries were analyzed for their linguistic characteristics. Finally, learners in a physics class reviewed the models, chose one, and explained their rationale.

Background

The role of AI in crafting expert models

With the advent of AI, Generative models, such as GPT-4 (OpenAI, 2023), have shown capabilities in producing human-like text. However, AI-generated content may suffer from issues like hallucinations and a lack of deep contextual understanding (Bender et al., 2021). These highlight the importance of combining human expertise with AI. Human-in-the-loop approaches utilize the strength of AI and human expertise (Shneiderman, 2020). In summarization, human experts can guide AI outputs, refining the content to enhance accuracy and relevance. The collaboration can potentially produce improved summaries. Mutual Theory of Mind (MTOM) suggests that humans and AI systems build models of each other's behaviors and intentions, improving communication (Wang & Goel, 2022). In the context of summarization, instructors can anticipate AI responses and adjust their inputs to optimize the generated summaries. Building on these, we developed a GenAI-enabled expert modeling tool

integrated within SMART. SMART supports learners in summarizing their understanding by concentrating on essential concepts and their relational structures (Kim, 2021; Kim et al., 2023a). It automates the analysis of linguistic inputs to help students build improved, deep concept maps, thereby enhancing their comprehension (Kim et al., 2023b). SMART uses multimodal feedback to show how closely learner responses align with an expert summary, with personalized messages aiding interpretation. Using GenAI, SMART guides students to a concept library where they can access clear definitions and specific examples that illustrate the connections between key ideas in propositional form. The expert modeling tool uses the original material in its input process, enabling GenAI to automatically generate a draft expert summary along with a list of identified key concepts. A human expert, typically the instructor, reviews the AI-generated model and chooses one of the three actions: 1) prompt AI to generate a new model, 2) add or remove key concepts and instruct AI to revise, 3) edit the summary directly, keeping the AI-proposed key concepts.

Characterizing the complexity of summaries

Writing is complex, integrating syntax, structure, semantics, grammar, and coherent idea organization (Hayes & Flower, 1980). Summarization intensifies this complexity by necessitating the extraction of key ideas. It requires higher-order cognitive skills, including comprehension, analysis, and synthesis (Kim et al., 2024). To evaluate complexities, linguistic performance is assessed using Complexity, Accuracy, and Fluency (CAF) measures (Skehan, 2009), alongside lexical complexity. Complexity reflects the writer's use of sophisticated syntactic structures (Ellis & Barkhuizen, 2005). Accuracy pertains to grammatical correctness, fluency involves the ease and flow of writing, and lexical complexity refers to the range and sophistication of vocabulary. This study also incorporates readability and semantic similarity. Readability measures how easily a text can be understood (Flesch, 1948). Effective summaries balance complexity, accuracy, and fluency to convey key information clearly and correctly. Semantic similarity examines how closely an expert summary corresponds to the source material. It ensures that the summarized content aligns with the intended knowledge outcomes (Kim et al., 2024).

Methods

The study included 41 undergraduates in an introductory physics course. Both classes were taught by a male instructor in the Department of Physics and Astronomy. The first expert modeling task, centered on the topic of "Velocity and Motion," was selected as the case. For this task, the instructor was asked to craft his own summary. The instructor used the expert modeling tool to design summary assignments and create individual expert summaries (up to 400 words). Three different expert models were subsequently prepared: the instructor-crafted, the initial AI-generated, and the Human-AI collaboration model. The summaries were analyzed using CAF measures: mean length of clause for complexity (Rivard & Gueye, 2023), error-free clause percentage for accuracy (Ellis & Yuan, 2004), and total word count for fluency (Plakans et al., 2019). The standardized number of different words for lexical complexity (Miller, 1991), the Flesch-Kincaid grade level for readability (Flesch, 1948), and both latent Dirichlet allocation (LDA) and latent semantic analysis (LSA) for semantic similarity (Crossley, 2020), were also used. A two-part survey gathered students' perceptions: first, ranking three summaries by how well they explained key concepts; second, a post-reflection to provide the rationale for their rankings. The log data included four actions: 1) regenerating: AI generates a new version of the summary; 2) viewing: the expert views the previous summaries; 3) post-editing: the expert refines the summary; and 4) modifying the prompt: the expert adds/removes key concepts, prompting AI to create a new summary.

Data analysis

The instructor's sequential interactions with the AI were analyzed using a descriptive, exploratory approach. A Friedman test was conducted to compare student rankings, followed by a post-hoc test to examine pairwise differences. A dual deductive-inductive approach was used for open-ended questions, with two researchers analyzing data and resolving differences by consensus. As a result, 141 instances were identified from 41 learner reflections and classified into four themes—Clarity (easy to understand), Depth (detailed information), Structure (balanced organization), and Coherence (logical connectedness)—each evaluated as either positive or negative.

Findings and discussion

Table 1 summarizes the measures across the three expert models. While based on a single case with minor differences, some notable variations were still observed. The Human model showed the lowest values in the CAF measures as well as lexical complexity and readability (FKGL). Its LSA value was highest (0.77), while LDA was similar to other models (0.96). Scholars have reported that human writers' cognitive capacity limits their attention to all linguistic components (Kellogg, 2014). Additionally, not every subject expert is a competent writer. For example, the instructor is a foreign, non-native English speaker, which may have led to reduced CAF. In the

same vein, the lexical complexity was the lowest, and FKGL indicated a high school level reading level (9.80). The human expert seemed to focus more on the main ideas, but there was no outstanding difference in semantic similarity from the AI-enabled models. AI and Human-AI collaboration models showed notable differences from the Human model across various measures, with the AI model exhibiting slightly more pronounced contrasts. This may be attributed to the AI's algorithmic. Similarly, the AI model showed the highest lexical complexity, likely from its algorithmic vocabulary selection. While the AI model performed well independently, the Human-AI collaboration slightly improved complexity and refined fluency.

Table 1

Linguistic Components of Expert Models

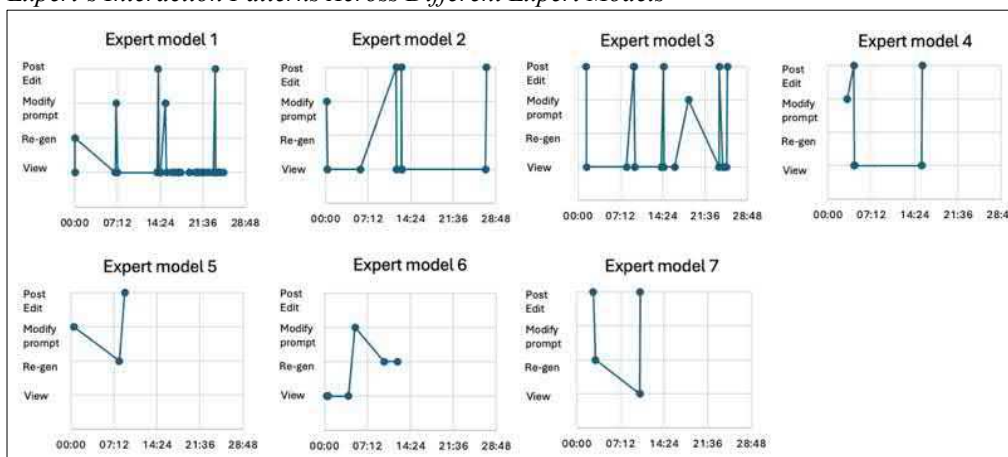
Expert Model	CAF Measures			Lexical Complexity ¹	Readability ²	Semantic similarity	
	Complexity	Accuracy	Fluency			LDA ³	LSA ⁴
Human	12.20	0.89	298	18.20	9.80	0.96	0.77
AI	14.50	1.00	468	19.00	12.00	0.95	0.76
Human - AI	14.80	1.00	378	18.60	12.00	0.97	0.72

Note. ¹Standardized Number of Different Words (NDWERZ) measure; ²Flesch-Kincaid Grade Level; ³Latent Dirichlet Allocation; ⁴Latent Semantic Analysis

Human Expert–AI interaction evolved toward greater procedural fluency and adaptive expertise (Wang & Goel, 2022), with early summaries being exploratory and later ones reflecting refined workflows (Figure 1).

Figure 1

Expert's Interaction Patterns Across Different Expert Models



The Friedman test indicated a statistically significant difference in the rankings of the three expert model types ($\chi^2(2, n = 41) = 6.16, p < 0.05$). Contrary to our expectations, the students' most preferred expert model was the AI's initial model (see Table 2). Post-hoc pairwise comparisons confirmed that the AI-generated summary was significantly preferred over the human-generated summary ($p < 0.05$).

Table 2

Results of Friedman Tests and Post-hoc Analyses

Expert Model	Rank 1	Rank 2	Rank 3	Average Rank	df	χ^2	p	post-hoc
Human	9	11	21	2.29	2	6.1605	0.04595	1<2 *
AI	15	20	6	1.78				1<3
Human – AI	17	11	13	1.90				2>3

Note. * $p < 0.05$. 1, 2, 3 indicate Human-generated, AI-generated, and Human-AI collaboration, respectively.

Thematic analysis results for open-ended questions focused on Clarity, Depth, Structure, and Coherence revealed that students valued the Human-generated summary for its clarity and simplicity. One student noted, "The model was easy to understand. It gave clear and concise definitions of the material we studied." This aligns

with the summary's highest LSA score (Table 1), suggesting effective communication of key concepts (Crossley, 2020). However, several students pointed out its limited depth and elaboration, consistent with lower scores in complexity and fluency (Ellis & Yuan, 2004; Skehan, 2009). In contrast, the AI-generated summary received the highest overall preference. It was frequently described as informative, well-structured, and cohesive. One student reflected, "*The model not only gives detailed explanations and key examples but includes differentiation between them and how to apply them.*" The Human-AI collaborative summary was recognized for its conceptual richness and instructional relevance. One student noted, "*It includes in-depth examples... which allows me to stay focused.*" However, some students found it "wordy" or "hard to follow," suggesting that the refinement process may have influenced structural coherence.

Conclusion

This study demonstrated the potential of GenAI to generate expert models, a process traditionally reserved for humans, by acting as a co-creator. Yet, it highlighted the crucial role of human experts in ensuring the quality of AI-generated outputs (Shneiderman, 2020). As a single case study, findings are not broadly generalizable and require careful contextualization. Also, the predominantly female student sample and the instructor's English fluency may have influenced the results. The nature of the learning materials, whether they pertain to open-ended or close-ended subject knowledge, may have impacted the ways expert summaries were generated. Further studies are necessary to explore these factors in greater depth.

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